

Unit 1. Vectors, Matrices, and the Data Matrix

Vectors in $\mathbb{R}^n$

Vectors, linear combinations, geometry

Definition 1.1 (Real n -vector). Let n be a positive integer. A **real n -vector** is an ordered list of n real numbers, written as a column,

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}, \quad x_i \in \mathbb{R}.$$

The number x_i is the **i th component** of $\mathbf{x}$. The set of all real n -vectors is written $\mathbb{R}^n$. Two vectors are equal when all their components are equal.

Origin: A&M p. 1

Definition 1.2 (Addition and scalar multiplication). For $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ and $\alpha \in \mathbb{R}$, define $\mathbf{x} + \mathbf{y}$ and

$\alpha \mathbf{x}$ componentwise:

$$(\mathbf{x} + \mathbf{y})_i = x_i + y_i, \quad (\alpha \mathbf{x})_i = \alpha x_i, \quad i = 1, \dots, n.$$

Write $\mathbf{0}$ for the vector with every component 0, and $-\mathbf{x}$ for $(-1)\mathbf{x}$.

Origin: A&M pp. 1–2

Theorem 1.3 (Arithmetic of vectors). For all $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{R}^n$ and all $\alpha, \beta \in \mathbb{R}$:

- (i) $\mathbf{x} + \mathbf{y} = \mathbf{y} + \mathbf{x}$;
- (ii) $(\mathbf{x} + \mathbf{y}) + \mathbf{z} = \mathbf{x} + (\mathbf{y} + \mathbf{z})$;
- (iii) $\mathbf{x} + \mathbf{0} = \mathbf{x}$ and $\mathbf{x} + (-\mathbf{x}) = \mathbf{0}$;
- (iv) $\alpha(\mathbf{x} + \mathbf{y}) = \alpha\mathbf{x} + \alpha\mathbf{y}$ and $(\alpha + \beta)\mathbf{x} = \alpha\mathbf{x} + \beta\mathbf{x}$;
- (v) $(\alpha\beta)\mathbf{x} = \alpha(\beta\mathbf{x})$ and $1\mathbf{x} = \mathbf{x}$.

Origin: Course formulation from A&M Ex. 1.3–1.5

Proof. For $i = 1, \dots, n$, the field axioms give

$$(\mathbf{x} + \mathbf{y})_i = x_i + y_i = y_i + x_i = (\mathbf{y} + \mathbf{x})_i,$$

$$((\mathbf{x} + \mathbf{y}) + \mathbf{z})_i = (x_i + y_i) + z_i = x_i + (y_i + z_i) = (\mathbf{x} + (\mathbf{y} + \mathbf{z}))_i,$$

$$(\mathbf{x} + \mathbf{0})_i = x_i + 0 = x_i,$$

$$(\mathbf{x} + (-\mathbf{x}))_i = x_i - x_i = 0,$$

$$(\alpha(\mathbf{x} + \mathbf{y}))_i = \alpha(x_i + y_i) = \alpha x_i + \alpha y_i = (\alpha\mathbf{x} + \alpha\mathbf{y})_i,$$

$$((\alpha + \beta)\mathbf{x})_i = (\alpha + \beta)x_i = \alpha x_i + \beta x_i = (\alpha\mathbf{x} + \beta\mathbf{x})_i,$$

$$((\alpha\beta)\mathbf{x})_i = (\alpha\beta)x_i = \alpha(\beta x_i) = (\alpha(\beta\mathbf{x}))_i,$$

$$(1\mathbf{x})_i = 1x_i = x_i.$$

Equality in every component gives each vector identity. □

Definition 1.4 (Linear combination). Given $\mathbf{x}_1, \dots, \mathbf{x}_k \in \mathbb{R}^n$ and $\alpha_1, \dots, \alpha_k \in \mathbb{R}$, the vector $\sum_{j=1}^k \alpha_j \mathbf{x}_j$ is a **linear combination** of $\mathbf{x}_1, \dots, \mathbf{x}_k$ with **coefficients** $\alpha_1, \dots, \alpha_k$.

Origin: S §1.1.1

Definition 1.5 (Two standard vectors). For $j = 1, \dots, n$, the **j th standard coordinate vector** $\mathbf{e}_j \in \mathbb{R}^n$ has j th component 1 and all other components 0. The vector $\mathbf{1} \in \mathbb{R}^n$ has every component equal to 1.

Origin: A&M p. 2

Theorem 1.6 (Coordinates). Every $\mathbf{x} \in \mathbb{R}^n$ satisfies $\mathbf{x} = \sum_{j=1}^n x_j \mathbf{e}_j$, and the coefficients in such a representation are unique: if $\sum_j \alpha_j \mathbf{e}_j = \sum_j \beta_j \mathbf{e}_j$ then $\alpha_j = \beta_j$ for every j .

Origin: A&M Ex. 1.13(b); S §1.1.1.1

Proof. For $i = 1, \dots, n$,

$$\left(\sum_{j=1}^n x_j \mathbf{e}_j \right)_i = \sum_{j=1}^n x_j (\mathbf{e}_j)_i = x_i.$$

If $\sum_j \alpha_j \mathbf{e}_j = \sum_j \beta_j \mathbf{e}_j$, comparison of component i gives $\alpha_i = \beta_i$. □

Example 1.1 (A linear combination). With $\mathbf{x} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$ and $\mathbf{y} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ in $\mathbb{R}^2$,

$$2\mathbf{x} - \mathbf{y} = \begin{pmatrix} 6 \\ 2 \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \end{pmatrix} = \begin{pmatrix} 5 \\ 0 \end{pmatrix}, \quad \mathbf{x} + \mathbf{y} = \begin{pmatrix} 4 \\ 3 \end{pmatrix}.$$

Origin: Course formulation

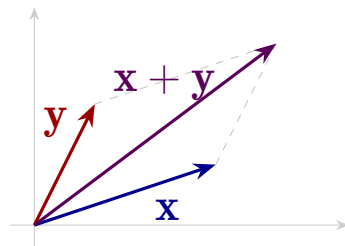


Figure 1.1: The two vectors of Example 1.1 and their sum.

1.2 Inner product, norm, and the Cauchy–Schwarz inequality

Definition 1.7 (Inner product). For $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ the **inner product** is the number

$$\mathbf{x}'\mathbf{y} = \sum_{i=1}^n x_i y_i.$$

Origin: A&M Ex. 1.7; S §1.3.5

Theorem 1.8 (Properties of the inner product). For all $\mathbf{x}, \mathbf{y}, \mathbf{z} \in \mathbb{R}^n$ and $\alpha \in \mathbb{R}$:

- (i) $\mathbf{x}'\mathbf{y} = \mathbf{y}'\mathbf{x}$;
- (ii) $(\alpha\mathbf{x} + \mathbf{z})'\mathbf{y} = \alpha(\mathbf{x}'\mathbf{y}) + \mathbf{z}'\mathbf{y}$, and likewise in the second argument;
- (iii) $\mathbf{x}'\mathbf{x} = \sum_i x_i^2 \geq 0$, with $\mathbf{x}'\mathbf{x} = 0$ if and only if $\mathbf{x} = \mathbf{0}$.

Origin: A&M Ex. 1.7

Proof.

$$\mathbf{x}'\mathbf{y} = \sum_i x_i y_i = \sum_i y_i x_i = \mathbf{y}'\mathbf{x},$$

$$(\alpha\mathbf{x} + \mathbf{z})'\mathbf{y} = \sum_i (\alpha x_i + z_i) y_i = \alpha \sum_i x_i y_i + \sum_i z_i y_i = \alpha\mathbf{x}'\mathbf{y} + \mathbf{z}'\mathbf{y}.$$

Symmetry gives linearity in the second argument. Finally,

$$\mathbf{x}'\mathbf{x} = \sum_i x_i^2 \geq 0, \quad \mathbf{x}'\mathbf{x} = 0 \iff x_i^2 = 0 \ (i = 1, \dots, n) \iff \mathbf{x} = \mathbf{0}.$$

□

Definition 1.9 (Norm). The **norm** of $\mathbf{x} \in \mathbb{R}^n$ is $\|\mathbf{x}\| = (\mathbf{x}'\mathbf{x})^{1/2}$. A vector with $\|\mathbf{x}\| = 1$ is a **unit vector**.

Origin: A&M Ex. 1.10; S §1.3.5

Theorem 1.10 (Basic properties of the norm). For $\mathbf{x} \in \mathbb{R}^n$ and $\alpha \in \mathbb{R}$,

$$\|\mathbf{x}\| \geq 0, \quad \|\mathbf{x}\| = 0 \iff \mathbf{x} = \mathbf{0}, \quad \|\alpha\mathbf{x}\| = |\alpha|\|\mathbf{x}\|.$$

Origin: A&M Ex. 1.10(a)–(b)

Proof. The first two claims follow from Theorem 1.8(iii). Moreover,

$$\|\alpha\mathbf{x}\|^2 = (\alpha\mathbf{x})'(\alpha\mathbf{x}) = \alpha^2\mathbf{x}'\mathbf{x} = |\alpha|^2\|\mathbf{x}\|^2.$$

Both sides of $\|\alpha\mathbf{x}\| = |\alpha|\|\mathbf{x}\|$ are nonnegative. □

Theorem 1.11 (Cauchy–Schwarz). For all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$,

$$|\mathbf{x}'\mathbf{y}| \leq \|\mathbf{x}\| \|\mathbf{y}\|,$$

with equality if and only if one of $\mathbf{x}, \mathbf{y}$ is a scalar multiple of the other.

Origin: A&M Ex. 1.9; S Prop. 1.5

Proof. For $\mathbf{y} = \mathbf{0}$, equality holds and $\mathbf{y} = 0\mathbf{x}$. Let $\mathbf{y} \neq \mathbf{0}$ and put $t = (\mathbf{x}'\mathbf{y})/(\mathbf{y}'\mathbf{y})$. Then

$$0 \leq \|\mathbf{x} - t\mathbf{y}\|^2 = \mathbf{x}'\mathbf{x} - 2t\mathbf{x}'\mathbf{y} + t^2\mathbf{y}'\mathbf{y} = \mathbf{x}'\mathbf{x} - \frac{(\mathbf{x}'\mathbf{y})^2}{\mathbf{y}'\mathbf{y}},$$

so

$$(\mathbf{x}'\mathbf{y})^2 \leq (\mathbf{x}'\mathbf{x})(\mathbf{y}'\mathbf{y}) = \|\mathbf{x}\|^2\|\mathbf{y}\|^2.$$

Taking nonnegative square roots gives the inequality. Equality is equivalent to $\|\mathbf{x} - t\mathbf{y}\| = 0$, hence to $\mathbf{x} = t\mathbf{y}$. Conversely, if $\mathbf{x} = \alpha\mathbf{y}$, then

$$|\mathbf{x}'\mathbf{y}| = |\alpha|\mathbf{y}'\mathbf{y} = |\alpha|\|\mathbf{y}\|^2 = \|\mathbf{x}\|\|\mathbf{y}\|$$

by Theorem 1.10. □

Corollary 1.12 (Triangle inequality). $\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|$ for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$.

Origin: A&M Ex. 1.10(c); S §1.3.5

Proof. Using Theorem 1.8 and then Theorem 1.11,

$$\|\mathbf{x} + \mathbf{y}\|^2 = \mathbf{x}'\mathbf{x} + 2\mathbf{x}'\mathbf{y} + \mathbf{y}'\mathbf{y} \leq \|\mathbf{x}\|^2 + 2\|\mathbf{x}\|\|\mathbf{y}\| + \|\mathbf{y}\|^2 = (\|\mathbf{x}\| + \|\mathbf{y}\|)^2.$$

Both sides before squaring are nonnegative. □

Definition 1.13 (Angle). For nonzero $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$, the **angle** between them is the unique $\theta \in [0, \pi]$ with

$$\cos \theta = \frac{\mathbf{x}'\mathbf{y}}{\|\mathbf{x}\| \|\mathbf{y}\|}.$$

Theorem 1.11 puts the right-hand side in $[-1, 1]$, and $\cos$ is a decreasing bijection from $[0, \pi]$ onto $[-1, 1]$, so exactly one such θ exists.

Origin: A&M Ex. 1.15

Definition 1.14 (Orthogonality). Vectors $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ are **orthogonal**, written $\mathbf{x} \perp \mathbf{y}$, when $\mathbf{x}'\mathbf{y} = 0$. By Definition 1.13 this is $\theta = \pi/2$ when both are nonzero, and $\mathbf{0}$ is orthogonal to everything.

Origin: A&M Ex. 1.12

Theorem 1.15 (Pythagoras). If $\mathbf{x} \perp \mathbf{y}$ then $\|\mathbf{x} + \mathbf{y}\|^2 = \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2$.

Origin: A&M Ex. 1.12(c)

Proof. $\|\mathbf{x} + \mathbf{y}\|^2 = \mathbf{x}'\mathbf{x} + 2\mathbf{x}'\mathbf{y} + \mathbf{y}'\mathbf{y}$ by Theorem 1.8, and the middle term is 0. $\square$

Example 1.2 (Orthogonality in $\mathbb{R}^3$). Let $\mathbf{x} = (1, 2, -1)'$ and $\mathbf{y} = (3, -1, 1)'$. Then $\mathbf{x}'\mathbf{y} = 3 - 2 - 1 = 0$, so $\mathbf{x} \perp \mathbf{y}$. Here $\|\mathbf{x}\|^2 = 6$, $\|\mathbf{y}\|^2 = 11$, and $\mathbf{x} + \mathbf{y} = (4, 1, 0)'$ has $\|\mathbf{x} + \mathbf{y}\|^2 = 17 = 6 + 11$.

Origin: Course formulation

1.3 The sample mean and variance as vector operations

Definition 1.16 (Sample mean and deviation vector). For $\mathbf{x} = (x_1, \dots, x_n)' \in \mathbb{R}^n$, the **sample mean** is $\bar{x} = n^{-1} \sum_i x_i$ and the **deviation vector** is $\mathbf{x} - \bar{x}\mathbf{1}$, whose i th component is $x_i - \bar{x}$.

Origin: Course formulation from A&M Ex. 2.37(b)

Theorem 1.17 (The mean as an inner product). $\bar{x} = n^{-1}\mathbf{1}'\mathbf{x}$, and $\mathbf{1}'\mathbf{1} = n$.

Origin: A&M Ex. 1.16, 2.37(b)

Proof.

$$\mathbf{1}'\mathbf{x} = \sum_i x_i = n\bar{x}, \quad \mathbf{1}'\mathbf{1} = \sum_i 1 = n.$$

□

Theorem 1.18 (The deviation vector is orthogonal to $\mathbf{1}$). $\mathbf{1}'(\mathbf{x} - \bar{x}\mathbf{1}) = 0$.

Origin: Adapted from A&M Ex. 2.37(b)

Proof. By Theorems 1.8 and 1.17,

$$\mathbf{1}'(\mathbf{x} - \bar{x}\mathbf{1}) = \mathbf{1}'\mathbf{x} - \bar{x}\mathbf{1}'\mathbf{1} = n\bar{x} - n\bar{x} = 0.$$

□

Definition 1.19 (Sample variance). The **sample variance** of $\mathbf{x}$ is

$$s_x^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2 = \frac{1}{n-1} \|\mathbf{x} - \bar{x}\mathbf{1}\|^2, \quad n \geq 2,$$

the second expression by Definitions 1.9 and 1.16. The **sample standard deviation** is $s_x = \sqrt{s_x^2}$.

Origin: Course formulation

Theorem 1.20 (Decomposition of the sum of squares). $\|\mathbf{x}\|^2 = n\bar{x}^2 + \|\mathbf{x} - \bar{x}\mathbf{1}\|^2$, that is $\sum_i x_i^2 = n\bar{x}^2 + \sum_i (x_i - \bar{x})^2$.

Origin: Course formulation

Proof. By Theorems 1.18, 1.15, and 1.17,

$$\begin{aligned}(\bar{x}\mathbf{1})'(\mathbf{x} - \bar{x}\mathbf{1}) &= \bar{x}\mathbf{1}'(\mathbf{x} - \bar{x}\mathbf{1}) = 0, \\ \|\mathbf{x}\|^2 &= \|\bar{x}\mathbf{1} + (\mathbf{x} - \bar{x}\mathbf{1})\|^2 \\ &= \|\bar{x}\mathbf{1}\|^2 + \|\mathbf{x} - \bar{x}\mathbf{1}\|^2 \\ &= n\bar{x}^2 + \|\mathbf{x} - \bar{x}\mathbf{1}\|^2.\end{aligned}$$

□

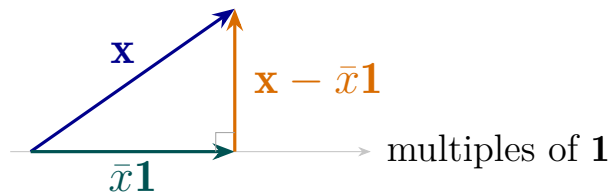


Figure 1.2: Theorem 1.20 in the plane containing $\mathbf{1}$ and $\mathbf{x}$.

Theorem 1.21 (The mean minimises the sum of squared deviations). For every $a \in \mathbb{R}$,

$$\sum_{i=1}^n (x_i - a)^2 = \sum_{i=1}^n (x_i - \bar{x})^2 + n(\bar{x} - a)^2,$$

so the left-hand side is smallest exactly at $a = \bar{x}$.

Origin: Course formulation

Proof. By Theorems 1.18, 1.15, and 1.17,

$$\begin{aligned}
 \sum_i (x_i - a)^2 &= \|\mathbf{x} - a\mathbf{1}\|^2 \\
 &= \|(\mathbf{x} - \bar{x}\mathbf{1}) + (\bar{x} - a)\mathbf{1}\|^2 \\
 &= \|\mathbf{x} - \bar{x}\mathbf{1}\|^2 + (\bar{x} - a)^2 \|\mathbf{1}\|^2 \\
 &= \sum_i (x_i - \bar{x})^2 + n(\bar{x} - a)^2.
 \end{aligned}$$

Since $n > 0$, $n(\bar{x} - a)^2 = 0 \iff a = \bar{x}$. □

Definition 1.22 (Sample covariance and correlation). For $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$ with $n \geq 2$, the **sample covariance** and, when both sample variances are nonzero, the **sample correlation** are

$$s_{xy} = \frac{(\mathbf{x} - \bar{x}\mathbf{1})'(\mathbf{y} - \bar{y}\mathbf{1})}{n - 1}, \quad r_{xy} = \frac{s_{xy}}{s_x s_y} = \frac{(\mathbf{x} - \bar{x}\mathbf{1})'(\mathbf{y} - \bar{y}\mathbf{1})}{\|\mathbf{x} - \bar{x}\mathbf{1}\| \|\mathbf{y} - \bar{y}\mathbf{1}\|}.$$

Origin: Course formulation

Theorem 1.23 (The correlation is a cosine). Whenever r_{xy} is defined, $r_{xy} = \cos \theta$, where θ is the angle between the deviation vectors $\mathbf{x} - \bar{x}\mathbf{1}$ and $\mathbf{y} - \bar{y}\mathbf{1}$. Consequently $|r_{xy}| \leq 1$, and $|r_{xy}| = 1$ if and only if

$$\mathbf{y} - \bar{y}\mathbf{1} = b(\mathbf{x} - \bar{x}\mathbf{1}) \quad \text{for some } b \neq 0,$$

that is, if and only if $y_i = a + bx_i$ for every i , with $a = \bar{y} - b\bar{x}$.

Origin: Course formulation

Proof. Put $\mathbf{u} = \mathbf{x} - \bar{x}\mathbf{1}$ and $\mathbf{v} = \mathbf{y} - \bar{y}\mathbf{1}$. Then

$$r_{xy} = \frac{\mathbf{u}'\mathbf{v}}{\|\mathbf{u}\|\|\mathbf{v}\|} = \cos \theta, \quad |r_{xy}| \leq 1.$$

By Theorem 1.11, equality holds iff $\mathbf{v} = b\mathbf{u}$ for some $b \neq 0$. Componentwise,

$$y_i - \bar{y} = b(x_i - \bar{x}) \iff y_i = (\bar{y} - b\bar{x}) + bx_i.$$

□

2 Matrices

2.1 Matrices, multiplication, transpose

Definition 1.24 (Matrix). An $m \times n$ **matrix** is a rectangular array $\mathbf{A} = (a_{ij})$ of real numbers with $i = 1, \dots, m$ indexing rows and $j = 1, \dots, n$ indexing columns. The set of all such is $\mathbb{R}^{m \times n}$. A matrix is **square** when $m = n$, and **of order** n then. Sums and scalar multiples are componentwise: $(\mathbf{A} + \mathbf{B})_{ij} = a_{ij} + b_{ij}$ and $(\alpha \mathbf{A})_{ij} = \alpha a_{ij}$. The matrix with every entry 0 is $\mathbf{O}$.

Origin: H §1.1, §1.2a–1.2b, §1.3a, §1.3f

Definition 1.25 (Product). For $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $\mathbf{B} \in \mathbb{R}^{n \times p}$ the product $\mathbf{AB} \in \mathbb{R}^{m \times p}$ has entries

$$(\mathbf{AB})_{ij} = \sum_{k=1}^n a_{ik} b_{kj}, \quad i = 1, \dots, m, \quad j = 1, \dots, p. \quad (1)$$

The product is defined only when the number of columns of $\mathbf{A}$ equals the number of rows of $\mathbf{B}$; the two matrices are then **conformable**.

Origin: H §1.2c

Definition 1.26 (Transpose). The **transpose** of $\mathbf{A} \in \mathbb{R}^{m \times n}$ is the matrix $\mathbf{A}' \in \mathbb{R}^{n \times m}$ with $(\mathbf{A}')_{ij} = a_{ji}$.

Origin: H §1.2d

Theorem 1.27 (Four readings of a product). Let $\mathbf{A} \in \mathbb{R}^{m \times n}$ have columns $\mathbf{a}_j$ and rows $\tilde{\mathbf{a}}'_i$, and let $\mathbf{B} \in \mathbb{R}^{n \times p}$ have columns $\mathbf{b}_j$ and rows $\tilde{\mathbf{b}}'_k$. Then

- (i) $(\mathbf{AB})_{ij} = \tilde{\mathbf{a}}'_i \mathbf{b}_j$;
- (ii) $\mathbf{AB} = (\mathbf{A}\mathbf{b}_1, \dots, \mathbf{A}\mathbf{b}_p)$, so the j th column of $\mathbf{AB}$ is $\mathbf{A}\mathbf{b}_j$;
- (iii) the i th row of $\mathbf{AB}$ is $\tilde{\mathbf{a}}'_i \mathbf{B}$;
- (iv) $\mathbf{AB} = \sum_{k=1}^n \mathbf{a}_k \tilde{\mathbf{b}}'_k$, a sum of n matrices each of the form (column)(row).

Origin: Course formulation from A&M Ex. 2.17, 5.2; H §1.2c, §2.2–2.3

Proof. For every i, j ,

$$\tilde{\mathbf{a}}'_i \mathbf{b}_j = \sum_k a_{ik} b_{kj} = (\mathbf{AB})_{ij},$$

$$(\mathbf{Ab}_j)_i = \sum_k a_{ik} (\mathbf{b}_j)_k = \sum_k a_{ik} b_{kj} = (\mathbf{AB})_{ij},$$

$$(\tilde{\mathbf{a}}'_i \mathbf{B})_j = \sum_k (\tilde{\mathbf{a}}_i)_k b_{kj} = \sum_k a_{ik} b_{kj} = (\mathbf{AB})_{ij},$$

$$\left(\sum_k \mathbf{a}_k \tilde{\mathbf{b}}'_k \right)_{ij} = \sum_k (\mathbf{a}_k)_i (\tilde{\mathbf{b}}_k)_j = \sum_k a_{ik} b_{kj} = (\mathbf{AB})_{ij}.$$



Corollary 1.28 (Matrix times vector). For $\mathbf{A} \in \mathbb{R}^{m \times n}$ with columns $\mathbf{a}_1, \dots, \mathbf{a}_n$ and $\mathbf{x} \in \mathbb{R}^n$,

$$\mathbf{Ax} = \sum_{j=1}^n x_j \mathbf{a}_j.$$

Origin: H §2.3

Proof. Theorem 1.27(iv), with $\mathbf{B} = \mathbf{x}$, gives $\mathbf{Ax} = \sum_k \mathbf{a}_k x_k$. □

Theorem 1.29 (Algebra of products). Whenever the products are defined,

- (i) $(\mathbf{AB})\mathbf{C} = \mathbf{A}(\mathbf{BC})$;
- (ii) $\mathbf{A}(\mathbf{B} + \mathbf{C}) = \mathbf{AB} + \mathbf{AC}$ and $(\mathbf{A} + \mathbf{B})\mathbf{C} = \mathbf{AC} + \mathbf{BC}$;
- (iii) $\alpha(\mathbf{AB}) = (\alpha\mathbf{A})\mathbf{B} = \mathbf{A}(\alpha\mathbf{B})$;
- (iv) $(\mathbf{A}')' = \mathbf{A}$, $(\mathbf{A} + \mathbf{B})' = \mathbf{A}' + \mathbf{B}'$, and $(\mathbf{AB})' = \mathbf{B}'\mathbf{A}'$.

Origin: Course formulation from A&M Ex. 2.4(b),(d), 2.14, 2.15(a); H §1.2c–1.2d, Ex. 1.2–1.3

Proof. (i) With $\mathbf{A} \in \mathbb{R}^{m \times n}$, $\mathbf{B} \in \mathbb{R}^{n \times p}$, and $\mathbf{C} \in \mathbb{R}^{p \times q}$, both sides are $m \times q$ and

$$((\mathbf{AB})\mathbf{C})_{ij} = \sum_{l=1}^p \left(\sum_{k=1}^n a_{ik} b_{kl} \right) c_{lj} = \sum_{k=1}^n \sum_{l=1}^p a_{ik} b_{kl} c_{lj} = \sum_{k=1}^n a_{ik} \left(\sum_{l=1}^p b_{kl} c_{lj} \right) = (\mathbf{A}(\mathbf{BC}))_{ij},$$

the middle step being the interchange of two finite sums.

(ii) For $\mathbf{B}, \mathbf{C} \in \mathbb{R}^{n \times p}$,

$$[\mathbf{A}(\mathbf{B} + \mathbf{C})]_{ij} = \sum_k a_{ik} (b_{kj} + c_{kj}) = (\mathbf{AB})_{ij} + (\mathbf{AC})_{ij}.$$

Similarly,

$$[(\mathbf{A} + \mathbf{B})\mathbf{C}]_{ij} = \sum_k (a_{ik} + b_{ik}) c_{kj} = (\mathbf{AC})_{ij} + (\mathbf{BC})_{ij}.$$

(iii)

$$[\alpha(\mathbf{AB})]_{ij} = \alpha \sum_k a_{ik} b_{kj} = \sum_k (\alpha a_{ik}) b_{kj} = \sum_k a_{ik} (\alpha b_{kj}).$$

(iv) For every i, j ,

$$((\mathbf{A}')')_{ij} = a_{ij}, \quad ((\mathbf{A} + \mathbf{B})')_{ij} = a_{ji} + b_{ji} = (\mathbf{A}' + \mathbf{B}')_{ij}.$$

For the product,

$$((\mathbf{AB})')_{ij} = (\mathbf{AB})_{ji} = \sum_k a_{jk} b_{ki} = \sum_k (\mathbf{B}')_{ik} (\mathbf{A}')_{kj} = (\mathbf{B}'\mathbf{A}')_{ij}. \quad \square$$

Definition 1.30 (Powers). For a square $\mathbf{A}$, define

$$\mathbf{A}^1 = \mathbf{A}, \quad \mathbf{A}^{p+1} = \mathbf{A}^p \mathbf{A} \quad (p \geq 1).$$

Thus $\mathbf{A}^2 = \mathbf{AA}$.

Origin: A&M p. 15

Example 1.3 (Multiplication does not commute). With $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$,

$$\mathbf{AB} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad \mathbf{BA} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

Moreover $\mathbf{A} \neq \mathbf{O}$ and $\mathbf{A}^2 = \mathbf{O}$.

Origin: Course formulation from A&M Ex. 2.8(b), 2.9(b)

Definition 1.31 (Identity matrix). $\mathbf{I}_n \in \mathbb{R}^{n \times n}$ has $(\mathbf{I}_n)_{ij} = 1$ when $i = j$ and 0 otherwise. Its columns are $\mathbf{e}_1, \dots, \mathbf{e}_n$. The subscript is dropped when the order is clear.

Origin: H §1.3d

Theorem 1.32. $\mathbf{A}\mathbf{I}_n = \mathbf{A}$ and $\mathbf{I}_m\mathbf{A} = \mathbf{A}$ for every $\mathbf{A} \in \mathbb{R}^{m \times n}$, and $\mathbf{I}'_n = \mathbf{I}_n$.

Origin: Adapted from A&M Ex. 2.18(b)

Proof.

$$(\mathbf{A}\mathbf{I}_n)_{ij} = \sum_k a_{ik}\delta_{kj} = a_{ij}, \quad (\mathbf{I}_m\mathbf{A})_{ij} = \sum_k \delta_{ik}a_{kj} = a_{ij}, \quad (\mathbf{I}'_n)_{ij} = \delta_{ji} = \delta_{ij}. \quad \square$$

2.2 The trace

Definition 1.33 (Trace). For a square $\mathbf{A}$ of order n , $\text{tr } \mathbf{A} = \sum_{i=1}^n a_{ii}$.

Origin: A&M Ex. 2.24; H §5.1

Theorem 1.34 (Elementary properties). For square $\mathbf{A}, \mathbf{B}$ of the same order and $\alpha, \beta \in \mathbb{R}$, $\text{tr}(\alpha\mathbf{A} + \beta\mathbf{B}) = \alpha \text{tr } \mathbf{A} + \beta \text{tr } \mathbf{B}$ and $\text{tr } \mathbf{A}' = \text{tr } \mathbf{A}$ and $\text{tr } \mathbf{I}_n = n$.

Origin: Adapted from A&M Ex. 2.24(a)–(c)

Proof.

$$\operatorname{tr}(\alpha \mathbf{A} + \beta \mathbf{B}) = \sum_i (\alpha a_{ii} + \beta b_{ii}) = \alpha \operatorname{tr} \mathbf{A} + \beta \operatorname{tr} \mathbf{B},$$

$$\operatorname{tr} \mathbf{A}' = \sum_i a_{ii} = \operatorname{tr} \mathbf{A},$$

$$\operatorname{tr} \mathbf{I}_n = \sum_i 1 = n.$$

□

Theorem 1.35 (The trace of a product). For $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $\mathbf{B} \in \mathbb{R}^{n \times m}$, both $\mathbf{AB}$ and $\mathbf{BA}$ are square, of orders m and n respectively, and

$$\operatorname{tr}(\mathbf{AB}) = \operatorname{tr}(\mathbf{BA}).$$

Origin: A&M Ex. 2.26(a); H §5.2

Proof. By Definitions 1.33 and 1.25,

$$\operatorname{tr}(\mathbf{AB}) = \sum_{i=1}^m (\mathbf{AB})_{ii} = \sum_{i=1}^m \sum_{k=1}^n a_{ik} b_{ki}, \quad \operatorname{tr}(\mathbf{BA}) = \sum_{k=1}^n (\mathbf{BA})_{kk} = \sum_{k=1}^n \sum_{i=1}^m b_{ki} a_{ik}.$$

The two double sums have the same terms, so they are equal. □

Corollary 1.36 (Cyclic property). $\operatorname{tr}(\mathbf{ABC}) = \operatorname{tr}(\mathbf{BCA}) = \operatorname{tr}(\mathbf{CAB})$ whenever the products are defined and square.

Origin: A&M Ex. 2.26(c); H §5.2

Proof. Apply Theorem 1.35 to the pair $\mathbf{A}$ and $\mathbf{BC}$, then to $\mathbf{B}$ and $\mathbf{CA}$, using associativity from Theorem 1.29(i). □

Theorem 1.37 (The trace as an inner product on matrices). For $\mathbf{A}, \mathbf{B} \in \mathbb{R}^{m \times n}$,

$$\operatorname{tr}(\mathbf{A}'\mathbf{B}) = \sum_{i=1}^m \sum_{j=1}^n a_{ij} b_{ij}.$$

In particular $\text{tr}(\mathbf{A}'\mathbf{A}) = \sum_{i,j} a_{ij}^2 \geq 0$, with $\text{tr}(\mathbf{A}'\mathbf{A}) = 0$ if and only if $\mathbf{A} = \mathbf{O}$.

Origin: A&M Ex. 2.24(d), 2.25, 2.38

Proof. $\mathbf{A}'\mathbf{B}$ is square of order n , and $(\mathbf{A}'\mathbf{B})_{jj} = \sum_i (\mathbf{A}')_{ji} b_{ij} = \sum_i a_{ij} b_{ij}$. Summing over j gives the stated double sum. Putting $\mathbf{B} = \mathbf{A}$ makes every term a square, so the sum is non-negative and vanishes only when every $a_{ij} = 0$. $\square$

Corollary 1.38. For $\mathbf{x} \in \mathbb{R}^n$, $\mathbf{x}'\mathbf{x} = \text{tr}(\mathbf{x}\mathbf{x}')$.

Origin: A&M Ex. 2.24(e)

Proof. By Theorem 1.35,

$$\text{tr}(\mathbf{x}\mathbf{x}') = \text{tr}(\mathbf{x}'\mathbf{x}) = \mathbf{x}'\mathbf{x},$$

because $\mathbf{x}'\mathbf{x}$ is a 1×1 matrix. $\square$

2.3 Special matrices

Definition 1.39 (Diagonal and triangular). A square $\mathbf{A}$ is **diagonal** when $a_{ij} = 0$ for all $i \neq j$; for $\mathbf{d} \in \mathbb{R}^n$, $\text{diag}(\mathbf{d})$ is the diagonal matrix with i th diagonal entry d_i . It is also written $\text{diag}(d_1, \dots, d_n)$. A square $\mathbf{A}$ is **upper triangular** when $a_{ij} = 0$ for $i > j$, and **lower triangular** when $a_{ij} = 0$ for $i < j$.

Origin: H §1.3c, §1.3g

Theorem 1.40. If $\mathbf{A}$ and $\mathbf{B}$ are upper triangular of order n , then $\mathbf{AB}$ is upper triangular and $(\mathbf{AB})_{ii} = a_{ii}b_{ii}$. The same holds for lower triangular matrices. In particular $\text{diag}(\mathbf{d})\text{diag}(\mathbf{c})$ is diagonal with i th diagonal entry $d_i c_i$.

Origin: Adapted from A&M Ex. 2.20, 2.21(b); S Ex. 3.1 (triangular closure)

Proof. $(\mathbf{AB})_{ij} = \sum_k a_{ik}b_{kj}$. A term is nonzero only if $k \geq i$ (else $a_{ik} = 0$) and $j \geq k$ (else $b_{kj} = 0$), hence only if $j \geq k \geq i$. If $i > j$ no such k exists and the entry is 0. If $i = j$ the only such k is $k = i$, and the term is $a_{ii}b_{ii}$. For lower triangular factors, a nonzero term requires $k \leq i$ and $j \leq k$, hence $j \leq k \leq i$; the same conclusions follow with the inequalities reversed. Diagonal matrices are both upper and lower triangular. $\square$

Definition 1.41 (Symmetric and skew-symmetric). A square $\mathbf{A}$ is **symmetric** when $\mathbf{A}' = \mathbf{A}$, and **skew-symmetric** when $\mathbf{A}' = -\mathbf{A}$.

Origin: H §1.3b, §14.6

Theorem 1.42 (Symmetric plus skew-symmetric). Every square $\mathbf{A}$ can be written as $\mathbf{A} = \mathbf{S} + \mathbf{K}$ with $\mathbf{S}$ symmetric and $\mathbf{K}$ skew-symmetric, in exactly one way, namely $\mathbf{S} = \frac{1}{2}(\mathbf{A} + \mathbf{A}')$ and $\mathbf{K} = \frac{1}{2}(\mathbf{A} - \mathbf{A}')$.

Origin: Adapted from A&M Ex. 2.23(b)

Proof. Those $\mathbf{S}$ and $\mathbf{K}$ sum to $\mathbf{A}$, and $\mathbf{S}' = \frac{1}{2}(\mathbf{A}' + \mathbf{A}) = \mathbf{S}$ while $\mathbf{K}' = \frac{1}{2}(\mathbf{A}' - \mathbf{A}) = -\mathbf{K}$ by Theorem 1.29(iv). For uniqueness, suppose $\mathbf{A} = \mathbf{S} + \mathbf{K}$ with $\mathbf{S}' = \mathbf{S}$ and $\mathbf{K}' = -\mathbf{K}$. Transposing gives $\mathbf{A}' = \mathbf{S} - \mathbf{K}$. Adding and subtracting the two equations gives $\mathbf{S} = \frac{1}{2}(\mathbf{A} + \mathbf{A}')$ and $\mathbf{K} = \frac{1}{2}(\mathbf{A} - \mathbf{A}')$. $\square$

Theorem 1.43. For any $\mathbf{A} \in \mathbb{R}^{m \times n}$, both $\mathbf{A}'\mathbf{A}$ and $\mathbf{A}\mathbf{A}'$ are symmetric.

Origin: Course formulation from A&M Ex. 2.22(c)

Proof.

$$(\mathbf{A}'\mathbf{A})' = \mathbf{A}'(\mathbf{A}')' = \mathbf{A}'\mathbf{A}, \quad (\mathbf{A}\mathbf{A}')' = \mathbf{A}(\mathbf{A}')' = \mathbf{A}\mathbf{A}'.$$

□

Definition 1.44 (Orthogonal matrix). A square $\mathbf{Q} = (\mathbf{q}_1, \dots, \mathbf{q}_n)$ is **orthogonal** when $\mathbf{Q}'\mathbf{Q} = \mathbf{I}_n$. Since $(\mathbf{Q}'\mathbf{Q})_{ij} = \mathbf{q}_i'\mathbf{q}_j$, this is equivalent to

$$\mathbf{q}_i'\mathbf{q}_j = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases}$$

Origin: Adapted from A&M Ex. 2.28(c)

Theorem 1.45 (Orthogonal invariance). If $\mathbf{Q}$ is orthogonal, then $(\mathbf{Q}\mathbf{x})'(\mathbf{Q}\mathbf{y}) = \mathbf{x}'\mathbf{y}$ and $\|\mathbf{Q}\mathbf{x}\| = \|\mathbf{x}\|$ for all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$. For nonzero $\mathbf{x}, \mathbf{y}$, the angles between $\mathbf{Q}\mathbf{x}, \mathbf{Q}\mathbf{y}$ and between $\mathbf{x}, \mathbf{y}$ are equal.

Origin: Course formulation from A&M Ex. 8.4(a)

Proof.

$$(\mathbf{Q}\mathbf{x})'(\mathbf{Q}\mathbf{y}) = \mathbf{x}'\mathbf{Q}'\mathbf{Q}\mathbf{y} = \mathbf{x}'\mathbf{y}, \quad \|\mathbf{Q}\mathbf{x}\|^2 = (\mathbf{Q}\mathbf{x})'(\mathbf{Q}\mathbf{x}) = \|\mathbf{x}\|^2.$$

The norms are nonnegative. For nonzero $\mathbf{x}, \mathbf{y}$,

$$\frac{(\mathbf{Q}\mathbf{x})'(\mathbf{Q}\mathbf{y})}{\|\mathbf{Q}\mathbf{x}\|\|\mathbf{Q}\mathbf{y}\|} = \frac{\mathbf{x}'\mathbf{y}}{\|\mathbf{x}\|\|\mathbf{y}\|}.$$

□

Definition 1.46 (Permutation matrix). A **permutation matrix** of order n is a square matrix whose entries are 0 and 1 with exactly one 1 in each row and exactly one 1 in each column.

Origin: A&M Ex. 2.29; H §8.4c

Theorem 1.47. Every permutation matrix $\mathbf{P}$ satisfies $\mathbf{P}'\mathbf{P} = \mathbf{P}\mathbf{P}' = \mathbf{I}_n$, so it is orthogonal.

Origin: A&M Ex. 2.29(d)

Proof. Write column j as $\mathbf{p}_j = \mathbf{e}_{\sigma(j)}$. The row condition makes σ bijective; hence

$$(\mathbf{P}'\mathbf{P})_{ij} = \mathbf{e}'_{\sigma(i)}\mathbf{e}_{\sigma(j)} = \begin{cases} 1, & i = j, \\ 0, & i \neq j. \end{cases}$$

Writing row i as $\mathbf{e}'_{\tau(i)}$ gives the same calculation for $\mathbf{P}\mathbf{P}'$. □

Definition 1.48 (Idempotent matrix). A square $\mathbf{A}$ is **idempotent** when $\mathbf{A}\mathbf{A} = \mathbf{A}$.

Origin: A&M Ex. 2.37; S Ex. 3.5 (projector condition)

Example 1.4 (The averaging matrix). Let $\mathbf{J} = n^{-1}\mathbf{1}\mathbf{1}' \in \mathbb{R}^{n \times n}$; $J_{ij} = 1/n$. Then

$$\mathbf{J}\mathbf{J} = n^{-2}\mathbf{1}(\mathbf{1}'\mathbf{1})\mathbf{1}' = n^{-2} \cdot n \cdot \mathbf{1}\mathbf{1}' = n^{-1}\mathbf{1}\mathbf{1}' = \mathbf{J},$$

and $\mathbf{J}' = \mathbf{J}$. For $\mathbf{x} \in \mathbb{R}^n$,

$$\mathbf{J}\mathbf{x} = n^{-1}\mathbf{1}(\mathbf{1}'\mathbf{x}) = \bar{x}\mathbf{1}.$$

Origin: Adapted from A&M Ex. 2.37(b)

3 The Data Matrix

3.1 Partitioned matrices and block multiplication

Definition 1.49 (Partition). Let $\mathbf{A} \in \mathbb{R}^{m \times n}$. Split the row indices $1, \dots, m$ into consecutive groups of sizes $m_1, \dots, m_r$ with $\sum_s m_s = m$, and the column indices into consecutive groups of sizes $n_1, \dots, n_c$ with $\sum_t n_t = n$. Writing $\mathbf{A}_{st}$ for the $m_s \times n_t$ submatrix cut out by group s of rows and group t of columns,

$$\mathbf{A} = \begin{pmatrix} \mathbf{A}_{11} & \cdots & \mathbf{A}_{1c} \\ \vdots & & \vdots \\ \mathbf{A}_{r1} & \cdots & \mathbf{A}_{rc} \end{pmatrix}$$

is a **partition** of $\mathbf{A}$ into blocks.

Origin: H §2.1

Theorem 1.50 (Block multiplication). Let $\mathbf{A} \in \mathbb{R}^{m \times n}$ be partitioned with column groups of sizes $n_1, \dots, n_c$, and let $\mathbf{B} \in \mathbb{R}^{n \times p}$ be partitioned with **row** groups of the same sizes $n_1, \dots, n_c$

and column groups of sizes $p_1, \dots, p_d$. Then every product $\mathbf{A}_{st}\mathbf{B}_{tu}$ below is defined and

$$(\mathbf{AB})_{su} = \sum_{t=1}^c \mathbf{A}_{st}\mathbf{B}_{tu}, \quad s = 1, \dots, r, \quad u = 1, \dots, d,$$

where $(\mathbf{AB})_{su}$ is the block of $\mathbf{AB}$ cut out by row group s and column group u .

Origin: Adapted from A&M Ex. 5.2; H §2.2

Proof. For i in row group s and j in column group u ,

$$(\mathbf{AB})_{ij} = \sum_{k=1}^n a_{ik}b_{kj} = \sum_{t=1}^c \sum_{k \in \text{group } t} a_{ik}b_{kj} = \left(\sum_{t=1}^c \mathbf{A}_{st}\mathbf{B}_{tu} \right)_{ij}.$$

□

Example 1.5 (Splitting off the constant column). Let $\mathbf{X} = (\mathbf{1}, \mathbf{X}_1) \in \mathbb{R}^{n \times (1+q)}$, where $\mathbf{X}_1$ holds

q variables. Applying Theorem 1.50 to $\mathbf{X}'\mathbf{X}$ with the same split on both sides,

$$\mathbf{X}'\mathbf{X} = \begin{pmatrix} \mathbf{1}' \\ \mathbf{X}_1' \end{pmatrix} (\mathbf{1}, \mathbf{X}_1) = \begin{pmatrix} \mathbf{1}'\mathbf{1} & \mathbf{1}'\mathbf{X}_1 \\ \mathbf{X}_1'\mathbf{1} & \mathbf{X}_1'\mathbf{X}_1 \end{pmatrix} = \begin{pmatrix} n & \mathbf{1}'\mathbf{X}_1 \\ \mathbf{X}_1'\mathbf{1} & \mathbf{X}_1'\mathbf{X}_1 \end{pmatrix},$$

using $\mathbf{1}'\mathbf{1} = n$ from Theorem 1.17. The off-diagonal block $\mathbf{1}'\mathbf{X}_1$ is the row of column sums of $\mathbf{X}_1$.

Origin: Course formulation

3.2 Gram matrix, covariance, correlation

Definition 1.51 (Data matrix). A **data matrix** is a matrix $\mathbf{X} \in \mathbb{R}^{n \times k}$ whose rows $\tilde{\mathbf{x}}_1', \dots, \tilde{\mathbf{x}}_n'$ hold n observations and whose columns $\mathbf{x}_1, \dots, \mathbf{x}_k$ hold k variables, so x_{ij} is the value of variable j on observation i .

Origin: Course formulation

Definition 1.52 (Gram matrix). The **Gram matrix** of $\mathbf{X}$ is $\mathbf{X}'\mathbf{X} \in \mathbb{R}^{k \times k}$.

Origin: Adapted from H Ex. 14.7

Theorem 1.53 (Two readings of the Gram matrix). $(\mathbf{X}'\mathbf{X})_{jl} = \mathbf{x}'_j\mathbf{x}_l$, and

$$\mathbf{X}'\mathbf{X} = \sum_{i=1}^n \tilde{\mathbf{x}}_i \tilde{\mathbf{x}}'_i.$$

The matrix $\mathbf{X}'\mathbf{X}$ is symmetric.

Origin: A&M Ex. 2.17; H §2.2, Ex. 14.7(a)

Proof. Theorem 1.27(i),(iv) gives

$$(\mathbf{X}'\mathbf{X})_{jl} = \mathbf{x}'_j\mathbf{x}_l, \quad \mathbf{X}'\mathbf{X} = \sum_i \tilde{\mathbf{x}}_i \tilde{\mathbf{x}}'_i.$$

Also $(\mathbf{X}'\mathbf{X})' = \mathbf{X}'\mathbf{X}$ by Theorem 1.43. □

Definition 1.54 (Centering matrix). $\mathbf{C} = \mathbf{I}_n - n^{-1}\mathbf{1}\mathbf{1}' \in \mathbb{R}^{n \times n}$.

Origin: A&M Ex. 2.37(b)

Theorem 1.55 (Properties of $\mathbf{C}$). (i) $\mathbf{C}' = \mathbf{C}$ and $\mathbf{C}\mathbf{C} = \mathbf{C}$;
(ii) $\mathbf{C}\mathbf{1} = \mathbf{0}$ and $\mathbf{1}'\mathbf{C} = \mathbf{0}'$;
(iii) $\mathbf{C}\mathbf{x} = \mathbf{x} - \bar{x}\mathbf{1}$ for every $\mathbf{x} \in \mathbb{R}^n$;
(iv) $\mathbf{C}\mathbf{X} = \mathbf{X} - \mathbf{1}\bar{\mathbf{x}}'$ for every $\mathbf{X} \in \mathbb{R}^{n \times k}$, where $\bar{\mathbf{x}} = n^{-1}\mathbf{X}'\mathbf{1}$, so $\mathbf{C}\mathbf{X}$ has all column means zero.

Origin: Adapted from A&M Ex. 2.37(b)

Proof. (i) Put $\mathbf{J} = n^{-1}\mathbf{1}\mathbf{1}'$. Then

$$\mathbf{C}' = \mathbf{I}' - \mathbf{J}' = \mathbf{I} - \mathbf{J} = \mathbf{C}, \quad \mathbf{C}\mathbf{C} = (\mathbf{I} - \mathbf{J})(\mathbf{I} - \mathbf{J}) = \mathbf{I} - 2\mathbf{J} + \mathbf{J}\mathbf{J} = \mathbf{I} - 2\mathbf{J} + \mathbf{J} = \mathbf{I} - \mathbf{J} = \mathbf{C}.$$

(ii)

$$\mathbf{C}\mathbf{1} = \mathbf{1} - n^{-1}\mathbf{1}(\mathbf{1}'\mathbf{1}) = \mathbf{0}, \quad \mathbf{1}'\mathbf{C} = (\mathbf{C}\mathbf{1})' = \mathbf{0}'.$$

(iii)

$$\mathbf{C}\mathbf{x} = \mathbf{x} - n^{-1}\mathbf{1}(\mathbf{1}'\mathbf{x}) = \mathbf{x} - \bar{x}\mathbf{1}.$$

(iv)

$$\mathbf{C}\mathbf{X} = (\mathbf{C}\mathbf{x}_1, \dots, \mathbf{C}\mathbf{x}_k) = \mathbf{X} - \mathbf{1}\bar{\mathbf{x}}', \quad \mathbf{1}'\mathbf{C}\mathbf{X} = \mathbf{0}'.$$

□

Definition 1.56 (Sample mean vector, covariance matrix, correlation matrix). For a data matrix $\mathbf{X} \in \mathbb{R}^{n \times k}$ with $n \geq 2$,

$$\bar{\mathbf{x}} = \frac{1}{n} \mathbf{X}' \mathbf{1} \in \mathbb{R}^k, \quad \mathbf{S} = \frac{1}{n-1} \mathbf{X}' \mathbf{C} \mathbf{X} \in \mathbb{R}^{k \times k}.$$

Write $s_j = \sqrt{s_{jj}}$. When every $s_{jj} > 0$, put $\mathbf{D}_s = \text{diag}(s_1, \dots, s_k)$ and define $\mathbf{R} = \mathbf{D}_s^{-1} \mathbf{S} \mathbf{D}_s^{-1}$, where $\mathbf{D}_s^{-1}$ is the diagonal matrix with j th diagonal entry $1/s_j$.

Origin: Course formulation from A&M Ex. 13.61 (mean vector only)

Theorem 1.57 ($\mathbf{S}$ is the matrix of sample covariances). $\mathbf{S} = (n-1)^{-1} (\mathbf{C} \mathbf{X})' (\mathbf{C} \mathbf{X})$, it is symmetric, and

$$s_{jl} = \frac{1}{n-1} \sum_{i=1}^n (x_{ij} - \bar{x}_j)(x_{il} - \bar{x}_l),$$

so s_{jl} is the sample covariance of variables j and l and $s_{jj} = s_j^2$ is the sample variance of variable j .

Origin: Course formulation

Proof.

$$(\mathbf{CX})'(\mathbf{CX}) = \mathbf{X}'\mathbf{C}'\mathbf{CX} = \mathbf{X}'\mathbf{CX} = (n-1)\mathbf{S}.$$

Thus $\mathbf{S}$ is symmetric. Its (j, l) entry is

$$s_{jl} = \frac{(\mathbf{x}_j - \bar{x}_j \mathbf{1})'(\mathbf{x}_l - \bar{x}_l \mathbf{1})}{n-1} = \frac{1}{n-1} \sum_i (x_{ij} - \bar{x}_j)(x_{il} - \bar{x}_l).$$

Definitions 1.19 and 1.22 give the stated identifications. □

Theorem 1.58 (Variance of a linear combination). For every $\mathbf{a} \in \mathbb{R}^k$, the derived variable $\mathbf{Xa} \in \mathbb{R}^n$ has sample variance $\mathbf{a}'\mathbf{Sa}$. Consequently $\mathbf{a}'\mathbf{Sa} \geq 0$ for every $\mathbf{a}$.

Origin: Course formulation

Proof. The deviation vector of $\mathbf{Xa}$ is $\mathbf{CXa}$. Hence its sample variance is

$$\frac{1}{n-1} \|(\mathbf{CX})\mathbf{a}\|^2 = \frac{1}{n-1} \mathbf{a}'(\mathbf{CX})'(\mathbf{CX})\mathbf{a} = \mathbf{a}'\mathbf{Sa}.$$

The first expression is nonnegative. □

Theorem 1.59 (Total variance). $\text{tr } \mathbf{S} = \sum_{j=1}^k s_j^2$, the sum of the sample variances of the k variables.

Origin: Course formulation

Proof.

$$\text{tr } \mathbf{S} = \sum_j s_{jj} = \sum_j s_j^2.$$

□

Theorem 1.60 (The correlation matrix). When every $s_{jj} > 0$, $r_{jl} = s_{jl}/(s_j s_l)$ is the sample correlation of variables j and l ; $r_{jj} = 1$; and $|r_{jl}| \leq 1$ for all j, l .

Origin: Course formulation

Proof.

$$r_{jl} = (\mathbf{D}_s^{-1} \mathbf{S} \mathbf{D}_s^{-1})_{jl} = \frac{s_{jl}}{s_j s_l}, \quad r_{jj} = \frac{s_{jj}}{s_j^2} = 1.$$

Theorem 1.23, applied to columns j, l of $\mathbf{X}$, gives $|r_{jl}| \leq 1$.

□

Example 1.6 (A data matrix of size 4×2). Let

$$\mathbf{X} = \begin{pmatrix} 1 & 3 \\ 2 & 4 \\ 3 & 8 \\ 6 & 9 \end{pmatrix}, \quad n = 4, \quad k = 2.$$

Then

$$\mathbf{X}'\mathbf{1} = \begin{pmatrix} 12 \\ 24 \end{pmatrix}, \quad \bar{\mathbf{x}} = \begin{pmatrix} 3 \\ 6 \end{pmatrix}, \quad \mathbf{CX} = \begin{pmatrix} -2 & -3 \\ -1 & -2 \\ 0 & 2 \\ 3 & 3 \end{pmatrix}.$$

$$(\mathbf{CX})'(\mathbf{CX}) = \begin{pmatrix} 14 & 17 \\ 17 & 26 \end{pmatrix}, \quad \mathbf{S} = \frac{1}{3} \begin{pmatrix} 14 & 17 \\ 17 & 26 \end{pmatrix}.$$

$$s_1^2 = \frac{14}{3}, \quad s_{12} = \frac{17}{3}, \quad s_2^2 = \frac{26}{3}, \quad r_{12} = \frac{17}{\sqrt{364}} \approx 0.891, \quad \text{tr } \mathbf{S} = \frac{40}{3}.$$

For $\mathbf{a} = (1, -1)'$,

$$\mathbf{Xa} = (-2, -2, -5, -3)', \quad \mathbf{a'Sa} = \frac{14 - 2(17) + 26}{3} = 2 = \frac{1}{3} \sum_{i=1}^4 ((\mathbf{Xa})_i + 3)^2.$$

Origin: Course formulation

4 Computation: Numerical Checks in R

```
1 set.seed(1)
2 X4 <- matrix(c(1, 2, 3, 6, 3, 4, 8, 9), nrow = 4) # small data example
3 data(Boston, package = "ISLR2")
4 XB <- as.matrix(Boston[, c("lstat", "rm", "age", "crim")])
5 n <- nrow(XB); k <- ncol(XB)
6 c(n = n, k = k)
```

```
##      n      k
## 506      4
```

4.1 The four readings of a product (§2.1)

```

1 A <- matrix(rnorm(12), 4, 3); B <- matrix(rnorm(15), 3, 5)
2 P      <- A %*% B
3 P_col <- sapply(1:5, function(j) A %*% B[, j])           # (ii) by columns of B
4 P_row <- t(sapply(1:4, function(i) A[i, ] %*% B))        # (iii) by rows of A
5 P_out <- Reduce(`+`, lapply(1:3, function(l)              # (iv) outer products
6                     outer(A[, l], B[l, ])))
7 c(columns = max(abs(P - P_col)), rows = max(abs(P - P_row)),
8   outer_products = max(abs(P - P_out)))

```

```

##           columns           rows outer_products
## 4.441e-16    4.441e-16    4.441e-16

```

```

1 xv <- rnorm(3)
2 c(combination = max(abs(A %*% xv -                          # Ax = sum x_j a_j
3   Reduce(`+`, lapply(1:3, function(j) xv[j] * A[, j])))),
4   transpose_rule = max(abs(t(A %*% B) - t(B) %*% t(A))))    # (AB)' = B'A'

```

```

## combination transpose_rule
##           0           0

```

4.2 The trace identities (§2.2)

```
1 M1 <- matrix(rnorm(12), 4, 3); M2 <- matrix(rnorm(12), 3, 4)
2 c(tr_M1M2 = sum(diag(M1 %*% M2)),          # order 4
3   tr_M2M1 = sum(diag(M2 %*% M1)),          # order 3
4   difference = sum(diag(M1 %*% M2)) - sum(diag(M2 %*% M1)))
```

```
##      tr_M1M2      tr_M2M1 difference
## -2.269e-01 -2.269e-01  5.551e-17
```

```
1 Aa <- matrix(rnorm(12), 4, 3); Bb <- matrix(rnorm(12), 4, 3)
2 c(tr_AtB = sum(diag(t(Aa) %*% Bb)), entrywise = sum(Aa * Bb),
3   tr_AtA = sum(diag(t(Aa) %*% Aa)), sum_of_squares = sum(Aa^2))
```

```
##          tr_AtB          entrywise          tr_AtA sum_of_squares
##        -0.6656        -0.6656         14.3774         14.3774
```

```
1 A3 <- matrix(rnorm(4), 2, 2); B3 <- matrix(rnorm(4), 2, 2)
2 C3 <- matrix(rnorm(4), 2, 2)
3 c(ABC = sum(diag(A3 %*% B3 %*% C3)), BCA = sum(diag(B3 %*% C3 %*% A3)),
4   CAB = sum(diag(C3 %*% A3 %*% B3)), ACB = sum(diag(A3 %*% C3 %*% B3)))
```

```
##      ABC      BCA      CAB      ACB
```

```
## 0.77203 0.77203 0.77203 0.05393
```

4.3 Centering, mean and variance (§1.3, §3.2)

```
1 n4 <- nrow(X4); C4 <- diag(n4) - matrix(1/n4, n4, n4)
2 c(idempotent = max(abs(C4 %*% C4 - C4)),           # CC = C
3   symmetric   = max(abs(C4 - t(C4))),              # C' = C
4   C_times_one = max(abs(C4 %*% rep(1, n4))),        # C1 = 0
5   vs_scale    = max(abs(C4 %*% X4 - scale(X4, scale = FALSE))))
```

```
## idempotent    symmetric C_times_one    vs_scale
##           0           0           0           0
```

```
1 S4 <- crossprod(C4 %*% X4) / (n4 - 1)           # S = X'CX/(n-1)
2 c(S11 = S4[1, 1], S12 = S4[1, 2], S22 = S4[2, 2], # 14/3, 17/3, 26/3
3   vs_cov = max(abs(S4 - cov(X4))),
4   r12     = S4[1, 2] / sqrt(S4[1, 1] * S4[2, 2]), # 0.891
5   r12_cor = cor(X4)[1, 2],
6   trace_S = sum(diag(S4)))                       # 40/3
```

```
##      S11      S12      S22 vs_cov      r12 r12_cor trace_S
## 4.667  5.667  8.667  0.000  0.891  0.891  13.333
```

```

1 av <- c(1, -1)
2 c(aSa = drop(t(av) %*% S4 %*% av), var_of_Xa = var(drop(X4 %*% av)))

```

```

##      aSa var_of_Xa
##      2         2

```

```

1 yv <- Boston$medv; nb <- length(yv)
2 c(sum_of_squares = sum(yv^2),
3   mean_part      = nb * mean(yv)^2,                                # n * xbar^2
4   deviation_part = sum((yv - mean(yv))^2),
5   difference     = sum(yv^2) - (nb * mean(yv)^2 + sum((yv - mean(yv))^2)))

```

```

## sum_of_squares      mean_part deviation_part      difference
##      299626         256910         42716              0

```

```

1 dx <- Boston$lstat - mean(Boston$lstat)
2 dy <- Boston$medv   - mean(Boston$medv)
3 cosine <- sum(dx * dy) / (sqrt(sum(dx^2)) * sqrt(sum(dy^2)))
4 c(cosine      = cosine, cor = cor(Boston$lstat, Boston$medv),
5   difference = cosine - cor(Boston$lstat, Boston$medv),

```

```
6 angle_in_degrees = acos(cosine) * 180 / pi)
```

```
##          cosine          cor          difference
##      -7.377e-01      -7.377e-01      1.110e-16
## angle_in_degrees
##      1.375e+02
```

```
1 CB <- diag(n) - matrix(1/n, n, n)
2 S  <- crossprod(CB %*% XB) / (n - 1)
3 Ds <- diag(1 / sqrt(diag(S)))
4 R  <- Ds %*% S %*% Ds
5 c(S_vs_cov = max(abs(S - cov(XB))), R_vs_cor = max(abs(R - cor(XB))),
6   trace_S  = sum(diag(S)), sum_of_variances = sum(apply(XB, 2, var)))
```

```
##          S_vs_cov          R_vs_cor          trace_S
##      2.160e-12      6.661e-16      9.178e+02
## sum_of_variances
##      9.178e+02
```


4.4 Association order and intermediate matrices

```

1 uu <- matrix(rnorm(1000), 1000, 1); vv <- matrix(rnorm(1000), 1, 1000)
2 ww <- matrix(rnorm(1000), 1000, 1)
3 tl <- system.time(for (i in 1:20) (uu %*% vv) %*% ww)[["elapsed"]] # 1000 x 1000
4 tr <- system.time(for (i in 1:20) uu %*% (vv %*% ww))[["elapsed"]] # scalar
5 c(max_abs_diff = max(abs((uu %*% vv) %*% ww - uu %*% (vv %*% ww))),
6   seconds_left = tl, seconds_right = tr)

```

```

## max_abs_diff seconds_left seconds_right
##      1.066e-13      3.320e-01      2.000e-03

```

```

1 m <- 3000
2 Y <- matrix(rnorm(m * 50), m, 50)
3 Cm <- diag(m) - matrix(1/m, m, m)
4 ta <- system.time(for (i in 1:5) Cm %*% Y)[["elapsed"]]
5 tb <- system.time(for (i in 1:5) Y - rep(1, m) %*% t(colMeans(Y)))[["elapsed"]]
6 c(max_abs_diff      = max(abs(Cm %*% Y - (Y - rep(1, m) %*% t(colMeans(Y))))),
7   megabytes_for_C    = as.numeric(object.size(Cm)) / 2^20,
8   seconds_via_matrix = ta, seconds_via_means = tb)

```

```

##      max_abs_diff      megabytes_for_C seconds_via_matrix

```

```
##          5.329e-15          6.866e+01          1.320e-01
## seconds_via_means
##          5.000e-03
```

```
1 Z <- matrix(rnorm(2000 * 300), 2000, 300)
2 t1 <- system.time(for (i in 1:20) t(Z) %*% Z)[["elapsed"]]
3 t2 <- system.time(for (i in 1:20) crossprod(Z))[["elapsed"]]
4 c(max_abs_diff = max(abs(t(Z) %*% Z - crossprod(Z))),
5   seconds_transpose_then_multiply = t1, seconds_crossprod = t2)
```

```
##          max_abs_diff
##          0.000
## seconds_transpose_then_multiply
##          0.179
##          seconds_crossprod
##          0.077
```

5 Pitfalls

5.1 Forming $\mathbf{C}$ to centre a large matrix

Explicit centering stores n^2 numbers. The run in §4.4 uses 68.7 MiB at $n = 3000$; 10^{10} double-precision entries at $n = 10^5$ use 80 GB. Compute $\mathbf{X} - \mathbf{1}\bar{\mathbf{x}}'$.

5.2 Reading $\mathbf{AB} = \mathbf{O}$ as $\mathbf{A} = \mathbf{O}$ or $\mathbf{B} = \mathbf{O}$

$$\mathbf{AA} = \mathbf{O} = \mathbf{AO}, \quad \mathbf{A} \neq \mathbf{O}$$

for the matrix $\mathbf{A}$ in Example 1.3; left cancellation fails.

```
1 A0 <- matrix(c(0, 0, 1, 0), 2, 2)           # nonzero, but A0 %*% A0 = 0
2 list(A0 = A0, A0_squared = A0 %*% A0)
```

```
## $A0
##      [,1] [,2]
## [1,]    0    1
## [2,]    0    0
##
## $A0_squared
##      [,1] [,2]
```

```
## [1,]    0    0
## [2,]    0    0
```

$$\mathbf{A} = \mathbf{0} \iff \text{tr}(\mathbf{A}'\mathbf{A}) = \sum_{i,j} a_{ij}^2 = 0.$$

```
1 c(tr_A0tA0 = sum(diag(t(A0) %*% A0)),      # nonzero, so A0 is not 0
2   tr_of_product = sum(diag(t(A0 %*% A0) %*% (A0 %*% A0))))
```

```
##      tr_A0tA0 tr_of_product
##              1              0
```

5.3 Two divisors, one name

For $n \geq 2$,

$$s_{n-1}^2 = \frac{1}{n-1} \sum_i (x_i - \bar{x})^2, \quad s_n^2 = \frac{1}{n} \sum_i (x_i - \bar{x})^2 = \frac{n-1}{n} s_{n-1}^2.$$

```
1 c(cov_default = cov(X4)[1, 1],              # divisor n - 1
2   divisor_n    = cov(X4)[1, 1] * (n4 - 1) / n4,
```

```
3 ratio = (n4 - 1) / n4
```

```
## cov_default divisor_n ratio
##      4.667      3.500    0.750
```

The divisor must be checked for each routine. It cancels from every correlation coefficient.

6 Exercises

Exercise 1.1. For $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$:

(a) prove the parallelogram identity

$$\|\mathbf{x} + \mathbf{y}\|^2 + \|\mathbf{x} - \mathbf{y}\|^2 = 2\|\mathbf{x}\|^2 + 2\|\mathbf{y}\|^2;$$

(b) deduce the polarization identity

$$\mathbf{x}'\mathbf{y} = \frac{1}{4} (\|\mathbf{x} + \mathbf{y}\|^2 - \|\mathbf{x} - \mathbf{y}\|^2) .$$

Origin: A&M Ex. 3.38(a); H Ex. 14.1 (polarization); S §1.3.4

Solution.

$$\|\mathbf{x} \pm \mathbf{y}\|^2 = \|\mathbf{x}\|^2 \pm 2\mathbf{x}'\mathbf{y} + \|\mathbf{y}\|^2.$$

Adding the $+$ and $-$ identities proves (a); subtracting them gives $\|\mathbf{x} + \mathbf{y}\|^2 - \|\mathbf{x} - \mathbf{y}\|^2 = 4\mathbf{x}'\mathbf{y}$, proving (b). 

Exercise 1.2. Determine exactly when equality holds in

$$\|\mathbf{x} + \mathbf{y}\| \leq \|\mathbf{x}\| + \|\mathbf{y}\|.$$

Include the cases $\mathbf{x} = \mathbf{0}$ and $\mathbf{y} = \mathbf{0}$.

Origin: Adapted from S Ex. 7.2

Solution. If $\mathbf{x} = \mathbf{0}$ or $\mathbf{y} = \mathbf{0}$, equality holds. Otherwise,

$$\begin{aligned} \|\mathbf{x} + \mathbf{y}\| = \|\mathbf{x}\| + \|\mathbf{y}\| &\iff \|\mathbf{x} + \mathbf{y}\|^2 = (\|\mathbf{x}\| + \|\mathbf{y}\|)^2 \\ &\iff \mathbf{x}'\mathbf{y} = \|\mathbf{x}\|\|\mathbf{y}\|. \end{aligned}$$

By Theorem 1.11, $\mathbf{x} = \alpha \mathbf{y}$ for some $\alpha \neq 0$. Then

$$\mathbf{x}'\mathbf{y} = \alpha \|\mathbf{y}\|^2, \quad \|\mathbf{x}\| \|\mathbf{y}\| = |\alpha| \|\mathbf{y}\|^2,$$

so equality holds iff $\alpha \geq 0$. Including the zero cases,

$$\|\mathbf{x} + \mathbf{y}\| = \|\mathbf{x}\| + \|\mathbf{y}\| \iff \exists \alpha \geq 0 : \quad \mathbf{x} = \alpha \mathbf{y} \text{ or } \mathbf{y} = \alpha \mathbf{x}.$$



Exercise 1.3. Show that $|\sum_i x_i| \leq \sqrt{n} \|\mathbf{x}\|$ for every $\mathbf{x} \in \mathbb{R}^n$, state when equality holds, and deduce that $|\bar{x}| \leq \|\mathbf{x}\|/\sqrt{n}$.

Origin: Adapted from A&M Ex. 12.4(a)

Solution.

$$|\sum_i x_i| = |\mathbf{1}'\mathbf{x}| \leq \|\mathbf{1}\| \|\mathbf{x}\| = \sqrt{n} \|\mathbf{x}\|.$$

Equality holds iff $\mathbf{x} = c\mathbf{1}$, that is, iff $x_1 = \cdots = x_n$. Division by n gives $|\bar{x}| \leq \|\mathbf{x}\|/\sqrt{n}$.



Exercise 1.4. (a) Exhibit matrices $\mathbf{A}, \mathbf{B}, \mathbf{C}$ of order 2 such that $\text{tr}(\mathbf{ABC}) \neq \text{tr}(\mathbf{ACB})$.
(b) If $\mathbf{A}, \mathbf{B}, \mathbf{C}$ are symmetric of the same order, prove $\text{tr}(\mathbf{ABC}) = \text{tr}(\mathbf{BAC})$.

Origin: Adapted from A&M Ex. 2.26(d); H Ex. 5.2

Solution. For (a), take

$$\mathbf{A} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad \mathbf{C} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

Then

$$\text{tr}(\mathbf{ABC}) = 1, \quad \text{tr}(\mathbf{ACB}) = 0.$$

For (b),

$$\text{tr}(\mathbf{ABC}) = \text{tr}((\mathbf{ABC})') = \text{tr}(\mathbf{CBA}) = \text{tr}(\mathbf{BAC}),$$

where the last equality is cyclic rotation. 

Exercise 1.5. Let $\mathbf{A}, \mathbf{B}$ be square matrices of the same order.

(a) Expand $(\mathbf{A} + \mathbf{B})^2$ and prove

$$(\mathbf{A} + \mathbf{B})^2 = \mathbf{A}^2 + 2\mathbf{AB} + \mathbf{B}^2 \iff \mathbf{AB} = \mathbf{BA}.$$

(b) If $\mathbf{A}$ and $\mathbf{B}$ are symmetric, prove

$$\mathbf{AB} \text{ is symmetric} \iff \mathbf{AB} = \mathbf{BA}.$$

(c) Give symmetric matrices of order 2 whose product is not symmetric.

Origin: Adapted from A&M Ex. 2.7(b),(e); H Ex. 1.5–1.6

Solution.

(a)

$$(\mathbf{A} + \mathbf{B})^2 = \mathbf{A}^2 + \mathbf{AB} + \mathbf{BA} + \mathbf{B}^2.$$

$$(\mathbf{A} + \mathbf{B})^2 = \mathbf{A}^2 + 2\mathbf{AB} + \mathbf{B}^2 \iff \mathbf{AB} + \mathbf{BA} = 2\mathbf{AB} \iff \mathbf{AB} = \mathbf{BA}.$$

(b)

$$(\mathbf{AB})' = \mathbf{B}'\mathbf{A}' = \mathbf{BA};$$

hence $\mathbf{AB} = (\mathbf{AB})'$ iff $\mathbf{AB} = \mathbf{BA}$.

(c) For

$$\mathbf{A} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad \mathbf{B} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix},$$

both factors are symmetric, whereas $\mathbf{AB} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ is not.

Exercise 1.6. Let $\mathbf{D} = \text{diag}(d_1, \dots, d_n)$ and $\mathbf{A} = (a_{ij}) \in \mathbb{R}^{n \times n}$.

(a) Prove $(\mathbf{DA})_{ij} = d_i a_{ij}$ and $(\mathbf{AD})_{ij} = a_{ij} d_j$.

(b) Prove

$$\mathbf{DA} = \mathbf{AD} \iff (d_i - d_j)a_{ij} = 0 \quad (1 \leq i, j \leq n).$$

(c) If $d_1, \dots, d_n$ are pairwise distinct, prove $\mathbf{DA} = \mathbf{AD}$ iff $\mathbf{A}$ is diagonal.

Origin: Adapted from A&M Ex. 2.11, 2.20

Solution. By Definition 1.25,

$$(\mathbf{DA})_{ij} = \sum_k d_{ik} a_{kj} = d_i a_{ij}, \quad (\mathbf{AD})_{ij} = \sum_k a_{ik} d_{kj} = a_{ij} d_j,$$

because only $k = i$ survives in the first sum and only $k = j$ in the second. Therefore

$$\mathbf{DA} = \mathbf{AD} \iff d_i a_{ij} = a_{ij} d_j \quad \forall i, j \iff (d_i - d_j) a_{ij} = 0 \quad \forall i, j.$$

If the d_i are pairwise distinct, then

$$i \neq j \implies d_i - d_j \neq 0 \implies a_{ij} = 0.$$

The converse follows because diagonal matrices commute by Theorem 1.40. ■

Exercise 1.7. For strictly upper triangular $\mathbf{A}$ of order n , prove by induction

$$(\mathbf{A}^p)_{ij} = 0 \quad (p \geq 1, j - i < p) \implies \mathbf{A}^n = \mathbf{O}.$$

Solution. For $p = 1$, $j - i < 1$ means $j \leq i$, so the claim is the hypothesis. Suppose it holds for p . A nonzero term in

$$(\mathbf{A}^{p+1})_{ij} = \sum_{k=1}^n (\mathbf{A}^p)_{ik} a_{kj}$$

requires

$$k - i \geq p, \quad j - k \geq 1, \quad j - i \geq p + 1.$$

Hence $(\mathbf{A}^{p+1})_{ij} = 0$ when $j - i < p + 1$. Since $j - i \leq n - 1 < n$ for every i, j , the case $p = n$ gives $\mathbf{A}^n = \mathbf{O}$. ■

Exercise 1.8. Let $\mathbf{K}$ be skew-symmetric of order n .

- (a) Prove $\mathbf{x}'\mathbf{K}\mathbf{x} = 0$ for every $\mathbf{x} \in \mathbb{R}^n$.
- (b) For arbitrary square $\mathbf{A}$, prove

$$\mathbf{x}'\mathbf{A}\mathbf{x} = \mathbf{x}' \left(\frac{\mathbf{A} + \mathbf{A}'}{2} \right) \mathbf{x}.$$

Solution. Put $q = \mathbf{x}'\mathbf{K}\mathbf{x} \in \mathbb{R}$. Then

$$q = q' = \mathbf{x}'\mathbf{K}'\mathbf{x} = -\mathbf{x}'\mathbf{K}\mathbf{x} = -q,$$

so $q = 0$. By Theorem 1.42,

$$\mathbf{A} = \frac{\mathbf{A} + \mathbf{A}'}{2} + \frac{\mathbf{A} - \mathbf{A}'}{2},$$

and the second summand is skew-symmetric. Part (a) gives the result. 

Exercise 1.9. Let $\mathbf{A}$ be square and satisfy $\mathbf{A}'\mathbf{A} = \mathbf{A}^2$. Prove that $\mathbf{A}$ is symmetric.

Origin: H Ex. 5.3

Solution. Put $\mathbf{D} = \mathbf{A} - \mathbf{A}'$. Then

$$\begin{aligned} \text{tr}(\mathbf{D}'\mathbf{D}) &= \text{tr}(\mathbf{A}'\mathbf{A}) - \text{tr}(\mathbf{A}'\mathbf{A}') - \text{tr}(\mathbf{A}^2) + \text{tr}(\mathbf{A}\mathbf{A}') \\ &= \text{tr}(\mathbf{A}^2) - \text{tr}((\mathbf{A}^2)') - \text{tr}(\mathbf{A}^2) + \text{tr}(\mathbf{A}'\mathbf{A}) \\ &= 0. \end{aligned}$$

Theorem 1.37 gives $\mathbf{D} = \mathbf{O}$, hence $\mathbf{A} = \mathbf{A}'$. 

Exercise 1.10. Let $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $\mathbf{B}, \mathbf{D} \in \mathbb{R}^{n \times p}$. Prove

$$\mathbf{AB} = \mathbf{0} \iff \mathbf{A}'\mathbf{AB} = \mathbf{0},$$

and deduce

$$\mathbf{AB} = \mathbf{AD} \iff \mathbf{A}'\mathbf{AB} = \mathbf{A}'\mathbf{AD}.$$

Origin: A&M Ex. 2.13(b)–(c)

Solution. The forward implication follows by premultiplication by $\mathbf{A}'$. Conversely,

$$\mathbf{A}'\mathbf{AB} = \mathbf{0} \implies (\mathbf{AB})'(\mathbf{AB}) = \mathbf{B}'\mathbf{A}'\mathbf{AB} = \mathbf{0} \implies \text{tr}((\mathbf{AB})'(\mathbf{AB})) = 0 \implies \mathbf{AB} = \mathbf{0}$$

by Theorem 1.37, applied to $\mathbf{AB}$. Replacing $\mathbf{B}$ by $\mathbf{B} - \mathbf{D}$ gives the second equivalence. ■

Exercise 1.11. Let $\mathbf{P} \in \mathbb{R}^{m \times r}$ satisfy $\mathbf{P}'\mathbf{P} = \mathbf{I}_r$, and put $\mathbf{H} = \mathbf{PP}'$.

- (a) Prove that $\mathbf{H}$ is symmetric and idempotent.
- (b) Prove that $\mathbf{Q} = \mathbf{I}_m - 2\mathbf{H}$ is orthogonal.

Origin: H Ex. 10.2–10.3

Solution.

$$\mathbf{H}' = \mathbf{P}\mathbf{P}' = \mathbf{H}, \quad \mathbf{H}^2 = \mathbf{P}(\mathbf{P}'\mathbf{P})\mathbf{P}' = \mathbf{P}\mathbf{P}' = \mathbf{H}.$$

Thus $\mathbf{Q}' = \mathbf{Q}$, and

$$\mathbf{Q}'\mathbf{Q} = (\mathbf{I}_m - 2\mathbf{H})^2 = \mathbf{I}_m - 4\mathbf{H} + 4\mathbf{H}^2 = \mathbf{I}_m.$$



Exercise 1.12. Let $\mathbf{P}$ and $\mathbf{Q}$ be idempotent matrices of the same order. Prove

$$(\mathbf{P} - \mathbf{Q})^2 + (\mathbf{I} - \mathbf{P} - \mathbf{Q})^2 = \mathbf{I}.$$

No commutativity assumption is permitted.

Origin: S Ex. 3.5(b)

Solution. Using $\mathbf{P}^2 = \mathbf{P}$ and $\mathbf{Q}^2 = \mathbf{Q}$ without changing the factor order,

$$(\mathbf{P} - \mathbf{Q})^2 = \mathbf{P} - \mathbf{P}\mathbf{Q} - \mathbf{Q}\mathbf{P} + \mathbf{Q},$$

$$(\mathbf{I} - \mathbf{P} - \mathbf{Q})^2 = \mathbf{I} - \mathbf{P} - \mathbf{Q} + \mathbf{P}\mathbf{Q} + \mathbf{Q}\mathbf{P}.$$

Their sum is $\mathbf{I}$. 

Exercise 1.13. Let

$$\mathbf{Z} = \begin{pmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{C} & \mathbf{D} \end{pmatrix},$$

where $\mathbf{A}$ and $\mathbf{D}$ are square. Prove:

- (a) $\mathbf{Z}$ is symmetric iff $\mathbf{A}, \mathbf{D}$ are symmetric and $\mathbf{C} = \mathbf{B}'$;
- (b) $\mathbf{Z}$ is diagonal iff $\mathbf{A}, \mathbf{D}$ are diagonal and $\mathbf{B} = \mathbf{O}, \mathbf{C} = \mathbf{O}$;
- (c) $\mathbf{Z}$ is upper triangular iff $\mathbf{A}, \mathbf{D}$ are upper triangular and $\mathbf{C} = \mathbf{O}$.

Origin: Adapted from A&M Ex. 5.3, 5.5

Solution. Definition 1.26 gives

$$\mathbf{Z}' = \begin{pmatrix} \mathbf{A}' & \mathbf{C}' \\ \mathbf{B}' & \mathbf{D}' \end{pmatrix}.$$

Hence

$$\mathbf{Z}' = \mathbf{Z} \iff \mathbf{A}' = \mathbf{A}, \quad \mathbf{D}' = \mathbf{D}, \quad \mathbf{C} = \mathbf{B}'.$$

For (b), every entry outside the two diagonal blocks must vanish, and each diagonal block must

itself be diagonal. For (c), every entry below the main diagonal consists of the entries below the diagonals of $\mathbf{A}, \mathbf{D}$ together with all entries of $\mathbf{C}$. ■

Exercise 1.14. For $\mathbf{B} \in \mathbb{R}^{m \times n}$ and $\mathbf{D} \in \mathbb{R}^{n \times m}$, put

$$\mathbf{Z} = \begin{pmatrix} -\mathbf{I}_m & \mathbf{B} \\ \mathbf{D} & \mathbf{I}_n \end{pmatrix}.$$

Prove

$$\mathbf{Z}^2 = \mathbf{I}_{m+n} \iff \mathbf{BD} = \mathbf{O} \text{ and } \mathbf{DB} = \mathbf{O}.$$

Give nonzero $\mathbf{B}, \mathbf{D}$ of order 2 satisfying both zero-product conditions.

Origin: Adapted from A&M Ex. 5.8

Solution. Block multiplication gives

$$\mathbf{Z}^2 = \begin{pmatrix} \mathbf{I}_m + \mathbf{BD} & -\mathbf{B} + \mathbf{B} \\ -\mathbf{D} + \mathbf{D} & \mathbf{I}_n + \mathbf{DB} \end{pmatrix} = \begin{pmatrix} \mathbf{I}_m + \mathbf{BD} & \mathbf{O} \\ \mathbf{O} & \mathbf{I}_n + \mathbf{DB} \end{pmatrix}.$$

For $m = n = 2$, take

$$\mathbf{B} = \mathbf{D} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \neq \mathbf{0}; \quad \mathbf{BD} = \mathbf{DB} = \mathbf{0}.$$

Exercise 1.15. Suppose

$$\mathbf{y} - \bar{y}\mathbf{1} = b(\mathbf{x} - \bar{x}\mathbf{1}), \quad b \neq 0,$$

and both sample variances are nonzero. Prove $r_{xy} = b/|b|$.

Origin: Course formulation

Solution. Put $\mathbf{u} = \mathbf{x} - \bar{x}\mathbf{1}$. Then

$$r_{xy} = \frac{\mathbf{u}'(b\mathbf{u})}{\|\mathbf{u}\| \|b\mathbf{u}\|} = \frac{b\|\mathbf{u}\|^2}{|b|\|\mathbf{u}\|^2} = \frac{b}{|b|}.$$

Thus $r_{xy} = 1$ for $b > 0$ and $r_{xy} = -1$ for $b < 0$.

Exercise 1.16. Let $\mathbf{X} = (\mathbf{1}, \mathbf{X}_1)$ as in Example 1.5. Prove

$$\mathbf{X}'\mathbf{C}\mathbf{X} = \begin{pmatrix} 0 & \mathbf{0}' \\ \mathbf{0} & \mathbf{X}_1'\mathbf{C}\mathbf{X}_1 \end{pmatrix}.$$

Origin: Course formulation from A&M Ex. 5.2

Solution. Theorem 1.55(ii) gives

$$\mathbf{C}\mathbf{X} = (\mathbf{C}\mathbf{1}, \mathbf{C}\mathbf{X}_1) = (\mathbf{0}, \mathbf{C}\mathbf{X}_1).$$

Therefore

$$\mathbf{X}'\mathbf{C}\mathbf{X} = \begin{pmatrix} \mathbf{1}' \\ \mathbf{X}_1' \end{pmatrix} (\mathbf{0}, \mathbf{C}\mathbf{X}_1) = \begin{pmatrix} 0 & \mathbf{1}'\mathbf{C}\mathbf{X}_1 \\ \mathbf{0} & \mathbf{X}_1'\mathbf{C}\mathbf{X}_1 \end{pmatrix},$$

and $\mathbf{1}'\mathbf{C} = \mathbf{0}'$. 

Exercise 1.17. Let $\mathbf{X} = (\mathbf{x}_1, \dots, \mathbf{x}_k) \in \mathbb{R}^{n \times k}$ and suppose

$$\mathbf{x}_k = c\mathbf{1} + \sum_{j=1}^{k-1} b_j \mathbf{x}_j.$$

Exhibit $\mathbf{a} \neq \mathbf{0}$ such that $\mathbf{a}'\mathbf{S}\mathbf{a} = 0$.

Origin: Course formulation

Solution. Take

$$\mathbf{a} = (b_1, \dots, b_{k-1}, -1)'$$

Then $\mathbf{a} \neq \mathbf{0}$ and

$$\mathbf{X}\mathbf{a} = \sum_{j=1}^{k-1} b_j \mathbf{x}_j - \mathbf{x}_k = -c\mathbf{1}, \quad \mathbf{a}'\mathbf{S}\mathbf{a} = \frac{1}{n-1} \|\mathbf{C}\mathbf{X}\mathbf{a}\|^2 = \frac{c^2}{n-1} \|\mathbf{C}\mathbf{1}\|^2 = 0$$

by Theorems 1.58 and 1.55(ii). 

Exercise 1.18. Let $\mathbf{P}$ be a permutation matrix of order n whose i th row is $\mathbf{e}'_{\tau(i)}$ and whose j th column is $\mathbf{e}_{\sigma(j)}$. For $\mathbf{Y} \in \mathbb{R}^{n \times p}$ and $\mathbf{Z} \in \mathbb{R}^{m \times n}$:

- (a) prove that row i of $\mathbf{PY}$ is row $\tau(i)$ of $\mathbf{Y}$;
- (b) prove that column j of $\mathbf{ZP}$ is column $\sigma(j)$ of $\mathbf{Z}$;
- (c) prove $\tau = \sigma^{-1}$.

Origin: Adapted from A&M Ex. 2.29(a)–(b)

Solution. Theorem 1.27(iii),(ii) gives

$$(\mathbf{PY})_{i\bullet} = \mathbf{e}'_{\tau(i)} \mathbf{Y} = \mathbf{Y}_{\tau(i)\bullet}, \quad (\mathbf{ZP})_{\bullet j} = \mathbf{Z} \mathbf{e}_{\sigma(j)} = \mathbf{Z}_{\bullet \sigma(j)}.$$

Moreover,

$$p_{ij} = 1 \iff j = \tau(i) \iff i = \sigma(j).$$

Hence $\sigma(\tau(i)) = i$ and $\tau(\sigma(j)) = j$. ■